Real Analysis: Limits of Functions

Limit Definition

Definition

Let $D \subset \mathbb{R}$ and $f : D \to \mathbb{R}$ be a function. Let c be a limit point of D and $l \in \mathbb{R}$. We say that $\lim_{x \to c} f(x) = l$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$|f(x) - l| < \epsilon \text{ for all } x \in N'(c, \delta) \cap D$$

Example

$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4$ since $|f(x) - 4| = |x - 2| < \epsilon$ when $|x - 2| < \epsilon$

Sequential Criterion

Theorem

Let $D \subset \mathbb{R}$, $f : D \to \mathbb{R}$, c a limit point of D, and $l \in \mathbb{R}$. Then:

$$\lim_{x \to c} f(x) = l \iff \textit{for every sequence } \{x_n\} \textit{ in } D - \{c\} \textit{ with } x_n \to c,$$

$$\textit{we have } f(x_n) \to l$$

Example

$\lim_{x \to 0} \sin \frac{1}{x}$ does not exist because:

- $x_n = \frac{2}{(4n-1)\pi} \to 0,\ f(x_n) = 1 \to 1$
- $y_n = \frac{1}{n\pi} \to 0,\ f(y_n) = 0 \to 0$

Uniqueness of Limits

Theorem

If a function $f : D \to \mathbb{R}$ *has a limit at a point* c*, then the limit is unique.*

Theorem (Boundedness)

Let $D \subset \mathbb{R}$ *and* $f : D \to \mathbb{R}$ *be a function. Let* c *be a limit point of* D*. If* $\lim_{x \to c} f(x) = l \in \mathbb{R}$*, then* f *is bounded on some neighbourhood of* c*.*

Proof

Let $\lim_{x \to c} f(x) = l$.

- Choose $\epsilon = 1$. Then $\exists\, \delta > 0$ such that

$$|f(x) - l| < 1 \quad \forall x \in N'(c, \delta) \cap D.$$

- Hence, $|f(x)| \leq |f(x) - l| + |l| < 1 + |l|$ for all $x \in N'(c, \delta) \cap D$.
- This shows f is bounded on $N(c, \delta) \cap D$ when $c \notin D$.

Proof (continued)

If $c \in D$, define

$$B = \max\{|f(c)| + 1, \; |l| + 1\}.$$

Then

$$|f(x)| < B \quad \forall x \in N(c, \delta) \cap D.$$

Thus f is bounded on $N(c, \delta) \cap D$.

Conclusion: In all cases, f is bounded on $N(c, \delta) \cap D$.

Corollary

If f is unbounded on every neighbourhood of c, then $\lim_{x\to c} f(x)$ does not exist in $\mathbb{R}$.

Example 1

$\lim_{x\to 0} \frac{1}{x}$ does not exist in $\mathbb{R}$.
Let $f(x) = \frac{1}{x}$, $x \in D = \mathbb{R} - \{0\}$. Here $0 \in D'$, and f is unbounded on every neighbourhood of 0. Thus, $\lim_{x\to 0} \frac{1}{x}$ does not exist in $\mathbb{R}$.

Algebra of Limits

Theorem

Let $D \subset \mathbb{R}$ and f and g are functions on D to $\mathbb{R}$. Let $c \in D'$ and $\lim_{x\to c} f(x) = l, \lim_{x\to c} g(x) = m$. *Then*

(i) $\lim_{x\to c}(f+g)(x) = l+m$,
where $f+g : D \to \mathbb{R}$ *is defined by* $(f+g)(x) = f(x)+g(x), x \in D$.

(ii) *if* $k \in \mathbb{R}$, $\lim_{x\to c}(kf)(x) = kl$,
where $kf : D \to \mathbb{R}$ *is defined by* $(kf)(x) = k.f(x)$, $x \in D$.

(iii) $\lim_{x\to c}(f.g)(x) = lm$,
where $f.g : D \to \mathbb{R}$ *is defined by* $(f.g)(x) = f(x).g(x), x \in D$.

(iv) $\lim_{x\to c} \frac{f}{g}(x) = \frac{l}{m}$,
where $\frac{f}{g} : D \to \mathbb{R}$ *is defined by* $\frac{f}{g}(x) = \frac{f(x)}{g(x)}, x \in D$, $m \neq 0$, *and* $g(x) \neq 0$ *for all* $x \in D$.

Sandwich Theorem

Let $D \subset \mathbb{R}$ and f, g, h be functions on D to $\mathbb{R}$. Let $c \in D'$. If $f(x) \leq h(x) \leq g(x)$ and $\lim f(x) = \lim g(x) = L$, then $\lim h(x) = L$.

Example:

- For $-1 \leq \sin x \leq 1$, dividing by x for $x > 0$ gives:

$$-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}.$$

Taking $x \to \infty$, both bounds $\to 0$. Hence,

$$\lim_{x \to \infty} \frac{\sin x}{x} = 0.$$

Proof

Let us choose $\epsilon > 0$.

- Since $\lim_{x \to c} f(x) = l$, there exists $\delta_1 > 0$ such that

$$l - \epsilon < f(x) < l + \epsilon \quad \forall x \in N'(c, \delta_1) \cap D.$$

- Since $\lim_{x \to c} h(x) = l$, there exists $\delta_2 > 0$ such that

$$l - \epsilon < h(x) < l + \epsilon \quad \forall x \in N'(c, \delta_2) \cap D.$$

Proof cont..

Let $\delta = \min\{\delta_1, \delta_2\}$.
Then for all $x \in N'(c, \delta) \cap D$, we have

$$l - \epsilon < f(x) < l + \epsilon, \quad l - \epsilon < h(x) < l + \epsilon.$$

Since $f(x) \leq g(x) \leq h(x)$, it follows that

$$l - \epsilon < g(x) < l + \epsilon \quad \forall x \in N'(c, \delta) \cap D.$$

Thus, $\lim_{x \to c} g(x) = l$.

Theorem (Product with Bounded Function)

Let $D \subset \mathbb{R}$ and f, g be functions on D to $\mathbb{R}$. Let $c \in D'$.
If f is bounded on $N'(c) \cap D$ for some deleted neighbourhood $N'(c)$ of c and $\lim\limits_{x \to c} g(x) = 0$, then

$$\lim_{x \to c} (f \cdot g)(x) = 0.$$

Example

$\lim_{x \to 0} x \cos \frac{1}{x} = 0$ and $\lim_{x \to 0} \sqrt{x} \sin \frac{1}{x} = 0$

Proof

Since f is bounded on $N'(c) \cap D$ for some neighbourhood $N(c)$ of c, there exists a number $B > 0$ and $\delta_1 > 0$ such that

$$|f(x)| < B \quad \forall x \in N'(c, \delta_1) \cap D.$$

Since $\lim_{x \to c} g(x) = 0$, $\exists\, \delta_2 > 0$ such that

$$|g(x) - 0| = |g(x)| < \tfrac{\epsilon}{B} \quad \forall x \in N'(c, \delta_2) \cap D.$$

Proof cont....

Let $\delta = \min\{\delta_1, \delta_2\}$.
Then for all $x \in N'(c, \delta) \cap D$,

$$|f(x)| < B, \quad |g(x)| < \tfrac{\epsilon}{B}.$$

Therefore,

$$|f(x)g(x)| \leq |f(x)|\,|g(x)| < B \cdot \tfrac{\epsilon}{B} = \epsilon.$$

Thus,

$$\lim_{x \to c} (f \cdot g)(x) = 0.$$

Example

An example when $f(x)$ is unbounded and $lim_{x \to c} g(x) = 0$ but $lim_{x \to c}(fg)(x)$ does not exist.

Consider

$$f(x) = \frac{1}{x}, \quad g(x) = x \sin\left(\tfrac{1}{x}\right), \quad x \neq 0.$$

- As $x \to 0$,

$$g(x) = x \sin\left(\tfrac{1}{x}\right) \to 0.$$

- But

$$(fg)(x) = \frac{1}{x} \cdot \big(x \sin(1/x)\big) = \sin(1/x).$$

- As $x \to 0$, $\sin(1/x)$ oscillates between -1 and 1, hence the limit does **not exist**.

$$\lim_{x \to 0} g(x) = 0, \quad f \text{ unbounded}, \quad \lim_{x \to 0} (fg)(x) \text{ does not exist.}$$

Infinite Limits

Definition

- $\lim_{x \to c} f(x) = \infty$ if for every $G > 0$, $\exists \delta > 0$ such that $f(x) > G$ for all $x \in N'(c, \delta) \cap D$
- $\lim_{x \to c} f(x) = -\infty$ if for every $G > 0$, $\exists \delta > 0$ such that $f(x) < -G$ for all $x \in N'(c, \delta) \cap D$

Example

$\lim_{x \to 0} \frac{1}{x^2} = \infty$ since for any $G > 0$, $\frac{1}{x^2} > G$ when $|x| < \frac{1}{\sqrt{G}}$

Sequential Criterion for Infinite Limits

Theorem

- $\lim_{x\to c} f(x) = \infty \iff$ *for every sequence* $\{x_n\}$ *in* $D - \{c\}$ *with* $x_n \to c$*, we have* $f(x_n) \to \infty$
- $\lim_{x\to c} f(x) = -\infty \iff$ *for every sequence* $\{x_n\}$ *in* $D - \{c\}$ *with* $x_n \to c$*, we have* $f(x_n) \to -\infty$

Example

$\lim_{x\to 0} \frac{1}{x}$ does not exist in $\mathbb{R}^*$ because:

- $x_n = \frac{1}{n} \to 0,\ f(x_n) = n \to \infty$
- $y_n = -\frac{1}{n} \to 0,\ f(y_n) = -n \to -\infty$

Limits at Infinity

Definition

- $\lim_{x \to \infty} f(x) = l$ if for every $\epsilon > 0$, $\exists G > 0$ such that $|f(x) - l| < \epsilon$ for all $x > G$
- $\lim_{x \to -\infty} f(x) = l$ if for every $\epsilon > 0$, $\exists G < 0$ such that $|f(x) - l| < \epsilon$ for all $x < G$

Example

$\lim_{x \to \infty} \frac{1}{x} = 0$ since for any $\epsilon > 0$, $\left|\frac{1}{x}\right| < \epsilon$ when $x > \frac{1}{\epsilon}$

Sequential Criterion for Limits at Infinity

Theorem

- $\lim_{x\to\infty} f(x) = l \iff$ *for every sequence* $\{x_n\}$ *with* $x_n \to \infty$, *we have* $f(x_n) \to l$
- $\lim_{x\to-\infty} f(x) = l \iff$ *for every sequence* $\{x_n\}$ *with* $x_n \to -\infty$, *we have* $f(x_n) \to l$

Example

$\lim_{x\to\infty} x \sin x$ does not exist because:

- $x_n = \frac{\pi}{2} + 2n\pi \to \infty$, $f(x_n) \to \infty$
- $y_n = -\frac{\pi}{2} + 2n\pi \to \infty$, $f(y_n) \to -\infty$

Trigonometric Limit

Result

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Idea.

Use geometric argument with sectors and Sandwich Theorem:

$$cosx < \frac{\sin x}{x} < 1 \text{ for } 0 < |x| < \frac{\pi}{2}$$

$\square$

Cauchy's Criterion

Theorem

Let $D \subset \mathbb{R}$ and $f : D \to \mathbb{R}$ be a function. Let c be a limit point of D. Then $\lim_{x \to c} f(x)$ exists if and only if for every $\epsilon > 0$, $\exists \delta > 0$ such that:

$$|f(x') - f(x'')| < \epsilon \text{ for all } x', x'' \in N'(c, \delta) \cap D$$

Example

$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \in \mathbb{R} - \mathbb{Q} \end{cases}$ has no limit anywhere because:

$$|f(x') - f(x'')| = 2 \text{ for any rational } x' \text{ and irrational } x''$$